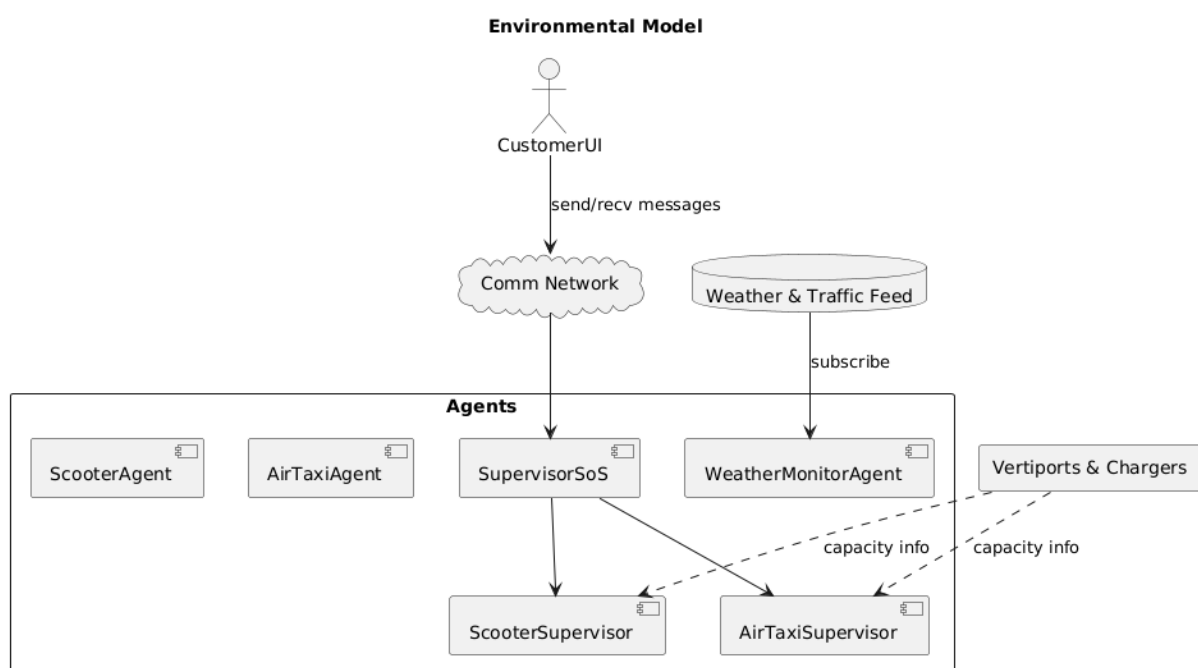


Designing MAS for Urban Air Mobility (UAM-SoS) Using Gaia Methodology

This Gaia-inspired design captures the organizational structure, roles, interactions, agent ↔ service mappings, and the environment in which the UAM-SoS MAS operates. It provides a clear blueprint for subsequent JADE (or other MAS framework) implementation.

1 Environment Model



Represents the abstract and computational environment where the MAS operates.

Entity	Type	Description and Constraints
Customer UI	Passive	Web/mobile interface where journey requests are submitted by user (start, destination, constraints like urgency, time limits).
Communication Network	Resource	Reliable message-passing middleware (e.g., JADE ACL over TCP); supports asynchronous communication.
Weather & Traffic Feed	Resource	External streams providing real-time data; agents subscribe for updates.
Vertiports & Chargers	Resource	Fixed infrastructure: capacity constraints (e.g., max scooters, recharging slots).
Electric Scooters	Active	Individual scooter agents (managed by S-CS1) with state: location, battery level, availability.

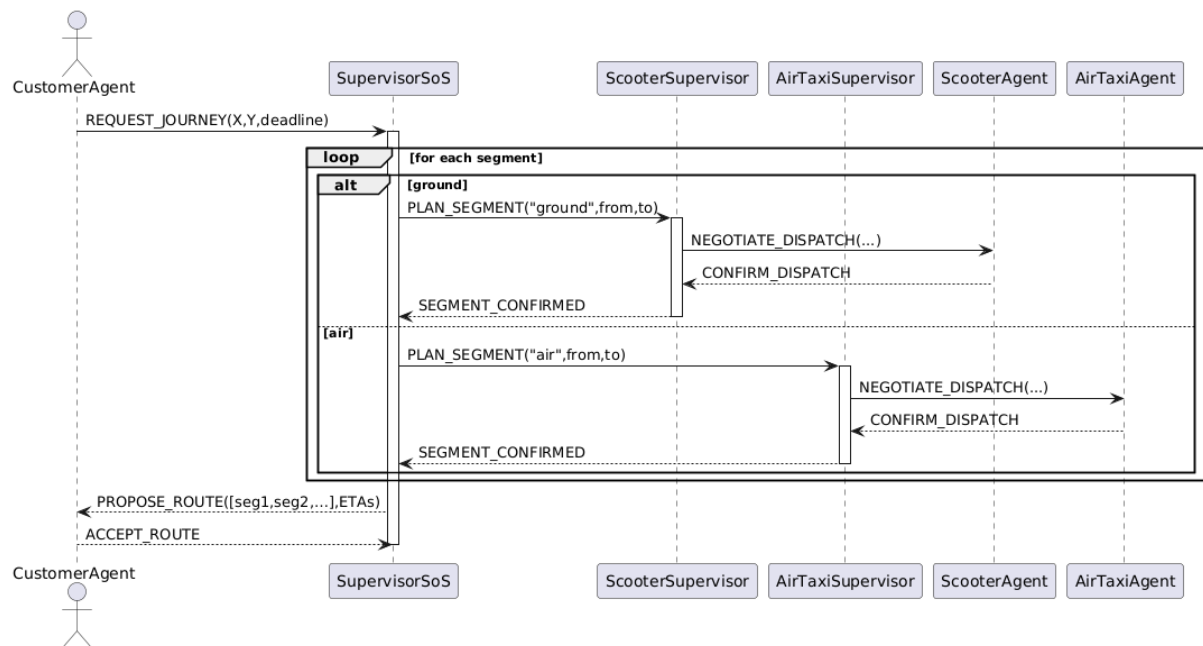
Air-Taxi Fleet	Active	Individual air-taxi agents (managed by S-CS2) with state: position, battery/fuel, flight corridors.
Supervisory Systems	Active	Holonic agents (S-SoS, S-CS1, S-CS2) coordinating subordinate resources.

Environmental Rules:

- **Resource Exclusivity:** A vehicle (scooter or taxi) can be assigned to at most one journey at a time.
- **Charging Constraints:** Scooter agents must reserve a charger before battery drops below threshold.
- **Airspace Safety:** Air-taxi agents must avoid no-fly zones and maintain separation minima.

2 Preliminary Role & Interaction Models

2.1 Scenario 1: On-Demand Intermodal Travel



2.1.1 Goal:

Get user from X→Y within T time duration.

2.1.2 Roles:

- CustomerAgent (CA): submits request, receives ETA & route
- SupervisorSoS (S-SoS): high-level planner, decomposer
- ScooterSupervisor (S-CS1): manages scooter network
- AirTaxiSupervisor (S-CS2): manages air-taxi fleet

2.1.3 Interactions (sequence overview)

- **CA → S-SoS:**
 - REQUEST_JOURNEY(X,Y,deadline)
- **S-SoS:**
 - Infers priority (e.g., medical)
 - Calls PLAN_SEGMENT("ground", X, vertiport_A) → S-CS1
 - Calls PLAN_SEGMENT("air", vertiport_A, vertiport_B) → S-CS2
 - Calls PLAN_SEGMENT("ground", vertiport_B, Y) → S-CS1
- **S-CS1 / S-CS2 → Resource Agents:**
 - NEGOTIATE_DISPATCH(vehicle_id, time_window)
- **Resource Agents → Supervisors:**
 - CONFIRM_DISPATCH(...) or REJECT(...)
- **S-SoS → CA:**
 - PROPOSE_ROUTE([scooter_A, taxi_B, scooter_C], ETAs)
- **CA → S-SoS:**
 - ACCEPT_ROUTE
- **All Supervisors → Vehicles:**
 - EXECUTE_JOURNEY(...)
- **Vehicles → CA (via supervisors):**
 - STATUS_UPDATE(location, ETA)

2.2 Scenario 2: Environmental Disruption Adaptation

2.2.1 Goal:

Reroute when sudden storm invalidates current plan.

2.2.2 Roles

Same as Scenario 1, plus:

- **WeatherMonitorAgent (WMA):** broadcasts disruption alerts

2.2.3 Interactions

- **WMA → All Supervisors:**
 - WEATHER_ALERT(storm_zone, severity)
- **S-CS1 / S-CS2:**
 - Receive alert, evaluate impact on assigned vehicles
 - Issue REQUEST_REPLAN(vehicle_id, new_constraints) to S-SoS
- **S-SoS:**
 - Adjust global plan: possibly swap modes, reassign vehicles
 - Issue updated segment plans to S-CS1/S-CS2
- **Supervisors → Vehicles:**

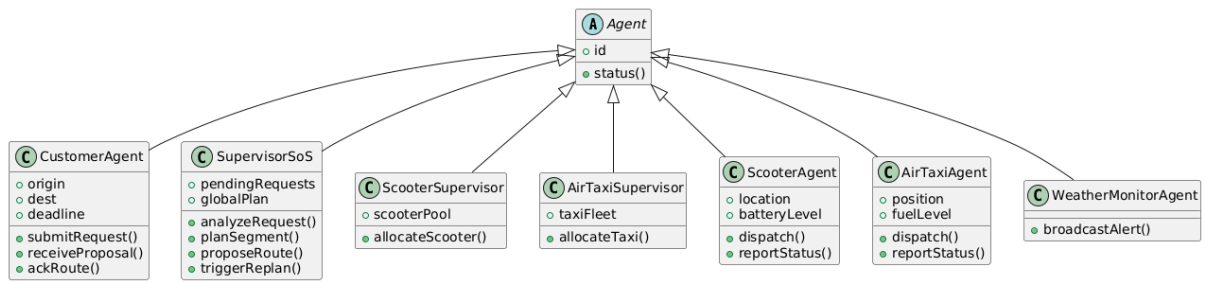
- UPDATE_ROUTE(new_route)
- **Vehicles → CA:**
 - Immediate STATUS_UPDATE("rerouted", new_ETAs)

3 Detailed Role Model: SupervisorSoS

Element	Description
Name	SupervisorSoS (S-SoS)
Description	Central coordinator for multi-modal journey planning, task decomposition, and adaptation to disruptions across subordinate systems.
Responsibilities	○ Decompose high-level journey into segments ○ Prioritize based on user constraints (e.g., deadlines, urgency) ○ Trigger replanning on alerts
Permissions	○ Read/write access to all subordinate agent status ○ Subscribe to weather/traffic feeds ○ Send dispatch/replan requests to S-CS1/S-CS2
Activities	○ analyzeRequest() ○ planSegment(mode, from, to) ○ aggregateSegmentPlans() ○ monitorAlerts() ○ triggerReplan()
Protocols	○ REQUEST_JOURNEY with CustomerAgent ○ PLAN_SEGMENT with S-CSx ○ WEATHER_ALERT subscription with WeatherMonitorAgent
Liveness	REQUEST_JOURNEY → PLAN_SEGMENT* → PROPOSE_ROUTE → (ACCEPT_ROUTE REJECT_ROUTE) S-SoS = (DecomposeRequest NegotiateResources HandleDisruption)*
Safety	○ Never propose a route violating battery or no-fly constraints ○ Always confirm subordinate plan feasibility before proposing to CustomerAgent ○ No overlapping air-taxi/scooter allocations. ○ Urgent requests (e.g., medical) prioritized.

4 Agent Model

Maps Gaia roles to concrete agent types, their state and services.



Agent Type	Implements Roles	State Variables	Services Offered
CustomerAgent	Customer	user_id, origin(X), dest(Y), deadline	submitRequest(), receiveProposal(), ackRoute()
SupervisorSoS	SupervisorSoS	pending_requests, global_plan, subscriptions	decomposeRequest(), monitor(), replan()
ScooterSupervisor	ScooterSupervisor	scooter_pool, charger_status, reservations	allocateScooter(), reallocateScooter()
AirTaxiSupervisor	AirTaxiSupervisor	taxi_fleet_status, vertiport_allocations	allocateTaxi(), rerouteTaxi()
ScooterAgent	(resource holon)	location, battery_level, status	dispatch(), reportStatus()
AirTaxiAgent	(resource holon)	position, fuel_level, flight_path	dispatch(), reportStatus()
WeatherMonitorAgent	Environment Monitor	weather_zones, severity_levels	broadcastAlert()

5 Service Model

Describes the externally visible services each agent offers, with signatures, pre-/post-conditions.

Service	Provider	Signature	Preconditions	Postconditions

REQUEST_JOURNEY	CustomerAgent	(user_id, X, Y, deadline)	UI inputs are valid locations; deadline $\geq$ travel_time_min	Creates a pending request in SupervisorSoS
PLAN_SEGMENT	SupervisorSoS	(mode, from, to, window)	mode $\in$ {ground, air}; environment model loaded	Dispatches NEGOTIATE_DISPATCH to sub-supervisor
NEGOTIATE_DISPATCH	S-CS1/S-CS2	(vehicle_type, time_window)	Vehicles are available; window fits operational constraints	Returns CONFIRM_DISPATCH or REJECT
PROPOSE_ROUTE	SupervisorSoS	([segments], ETAs)	All segments confirmed	CustomerAgent can accept or reject
EXECUTE_JOURNEY	All Supervisors	(route_plan_id)	Route accepted by customer	Vehicles start execution; status streaming begins
WEATHER_ALERT	WeatherMonitorAgent	(zone, severity, timestamp)	Connected subscribers	Supervisory agents may trigger triggerReplan()
triggerReplan	SupervisorSoS	(affected_segment_ids)	Alert indicates disruption on active segment	Revised segment plans are sent to S-CS1/S-CS2
REPORT_STATUS	Resource Agents	(agent_id, location, ETA, battery/fuel)	Agent is en route	Supervisors and customer receive live updates